

CStats - Lab #4

March 17, 2026

Exercise 1.

A computer ANOVA output is shown in Table 1. Fill in the blanks. You may give bounds on the p -value.

One-way ANOVA					
Source	DF	SS	MS	F	p
Factor	3	36.15	?	?	?
Error	?	?	?		
Total	19	196.04			

Table 1: One-way ANOVA Table

Exercise 2.

The engineer is interested in determining if the RF power setting affects the etch rate, and she has run a completely randomized experiment with four levels of RF power and five replicates. The data are shown in Table 2. Construct the ANOVA table and test the null hypothesis $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$ against the alternative H_1 : *some means are different*, for the significance level $\alpha = 0.05$.

RF Power (W)	Observed Etch Rate ($\text{\AA}/\text{min}$)					Totals	Averages
	1	2	3	4	5	$y_{i.}$	$\bar{y}_{i.}$
160	575	542	530	539	570	2756	551.2
180	565	593	590	579	610	2937	587.4
200	600	651	610	637	629	3127	625.4
220	725	700	715	685	710	3535	707.0
						$y_{..} = 12,355$	$\bar{y}_{..} = 617.75$

Table 2: The Plasma Etching Experiment

Exercise 3.

Let X_1, X_2, X_3, X_4 equal the cholesterol level of a woman under the age of 50, a man under 50, a woman 50 or older, and a man 50 or older, respectively. Assume that the distribution of X_i is $N(\mu_i, \sigma^2)$, $i = 1, 2, 3, 4$.

(a) Define null and alternative hypotheses and give a critical region for an $\alpha = 0.05$ significance level.

(b) Construct an ANOVA table and state your conclusion, using the following data:

X_1 :	221	213	202	183	185	197	162
X_2 :	271	192	189	209	227	236	142
X_3 :	262	193	224	201	161	178	265
X_4 :	192	253	248	278	232	267	289

(c) Give bounds on the p -value for this test.

Exercise 4.

A pharmaceutical manufacturer wants to investigate the bioactivity of a new drug. A completely randomized single-factor experiment was conducted with three dosage levels, and the following results were obtained:

Dosage	Observations
20g	24 28 37 30
30g	37 41 31 35
40g	42 47 52 38

(a) Is there evidence to indicate that dosage level affects bioactivity? Use $\alpha = 0.05$.

(b) Analyze the residuals from this experiment and comment on model adequacy.